

ROHAAN AHMED

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SUMMARY

Aspiring **Full-Stack Developer** with a deep passion for **Web Development** and **Software Engineering**.

Currently pursuing a Bachelor of Computer Science (BCS) and boasting a practical and proficient skill set in both backend and frontend technologies. Experienced in deploying robust applications using modern tech stacks and methodologies. Seeking a dynamic role as a Backend, Frontend, or Full-Stack Engineer where I can leverage my skills to contribute to innovative and impactful projects.

EDUCATION

University of Southern Queensland, Australia

Bachelors of ICT (Data Science and Artificial Intelligence & App Development)

Expected Graduation: December 2025

- GPA: 6.6/7.0

EXPERIENCE

Fullstack Software Engineering Intern | Energy Queensland Advanced Analytics | Nov 2024 – Feb 2025

Led development of a geospatial application for statewide electrical infrastructure inspection, with full ownership of design and implementation:

- Engineered React/TypeScript frontend with Leaflet maps integration, enabling intuitive infrastructure data querying while maintaining <40MB package size
- Developed FastAPI backend with H3/GeoJSON processing and Redis caching, optimizing geospatial queries for thousands of infrastructure assets
- Implemented Three.js-based LiDAR visualization system with intelligent denoising for classified Laz/Parquet files
- Drove full development lifecycle using CI/CD, Jira, and Confluence, delivering features through structured sprints based on user feedback

SKILLS

- **Languages & Frameworks:** Python, JavaScript/TypeScript, Java, C++
- **Frontend Technologies:** React, Tailwind CSS, SASS, CSS
- **Backend & APIs:** Django, FastAPI, Flask, Node.js, SpringBoot
- **Databases:** PostgreSQL, MongoDB, DuckDB
- **DevOps & Deployment:** Docker, Kubernetes, Git, CircleCI, GitHub Actions, DigitalOcean, AWS
- **Tools & Platforms:** Postman, Redis, RabbitMQ, Kafka, WebSocket
- **Software Development Practices:** Object-Oriented Programming, Write-Through and Write-Behind Caching, Linux Environments

SELECTED PROJECTS

1. [NucleiSystems](#) | Digital Asset Storage Platform
 - Built custom IPFS-Kubo binary bindings to replace abandoned official library, enabling robust asset management
 - Engineered self-managing filesystem for automated asset fetching and cleaning
 - Implemented multi-layer caching architecture with Redis and secondary database for optimized asset retrieval
 - Developed sustainable backend architecture overcoming framework limitations for scalable asset storage
2. [r3almX](#) | Real-Time Communications Platform
 - Engineered high-throughput WebSocket system with RabbitMQ for targeted room message broadcasting
 - Developed separate WebSocket connection for custom push-notifications and event management
 - Implemented dynamic table generation with custom ORM modifications for flexible data modeling
 - Created extensible plugin system for frontend room customization and feature modification
 - Optimized performance using asynchronous task threads for message broadcasting and database operations
 - Developed parallel backend implementations in both Python and Java for performance comparison
3. [BlockFrame](#) | File Chunking Library
 - Developed ORM-based asset management system with customizable table models for varied storage requirements
 - Implemented Façade pattern for simplified integration across different use cases
 - Created vectorized search system with database caching for optimized file chunk retrieval
 - Built highly configurable chunking system with comprehensive settings management